Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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The question

A decoder fit to one animal's cortex **drifts**: it loses accuracy on that same animal weeks later. The standard remedy is recalibration — collecting fresh labelled data from the subject, over and over.

Population structure is known to be *shared* across individuals, and pooled decoders work. But every one of those results is measured **at a single moment in time**.

Nobody had asked whether pooling resists drift.

The prediction

A decoder that *never saw* the subject cannot overfit that subject's drift-prone idiosyncrasies — so it should age more slowly. That difference is measurable as an **asymmetry** between the two arms.

Setup

BraiDyn-BC: 25 mice, 44 Allen-atlas cortical parcels, widefield calcium, ∼15 daily sessions over 3 weeks. Four task events decoded against matched negatives (ℓ_2 logistic regression, AUC).

Per mouse, compare **aging** — accuracy lost from an early block (days 1–5) to a late block (days 11–15), training held fixed — for a decoder trained on **itself** versus one **pooled over the other 24 mice**.

Whole sessions are held out, not trials. Only the within-subject arm can share a recording session between train and test, so a trial-level split inflates *that arm alone*.

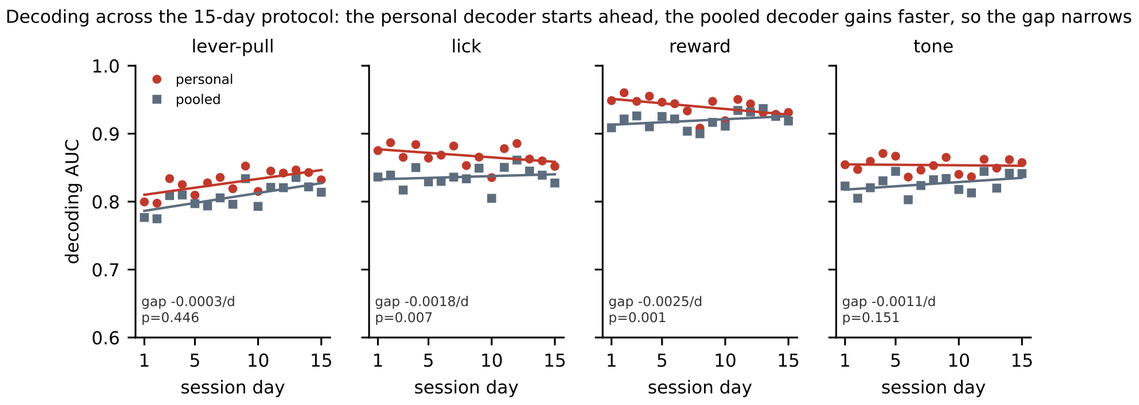
The effect

$+0.017$

AUC the personal decoder ages *more*

$p = 5 \times 10^{-5}$ · 73 of 100 mouse–event pairs

A day-by-day regression over all fifteen sessions agrees with the two-block contrast: the personal decoder's early lead of ∼0.035 AUC narrows by 0.0014/day ($p = 5 \times 10^{-5}$), so **60% of it is gone by day 15**.



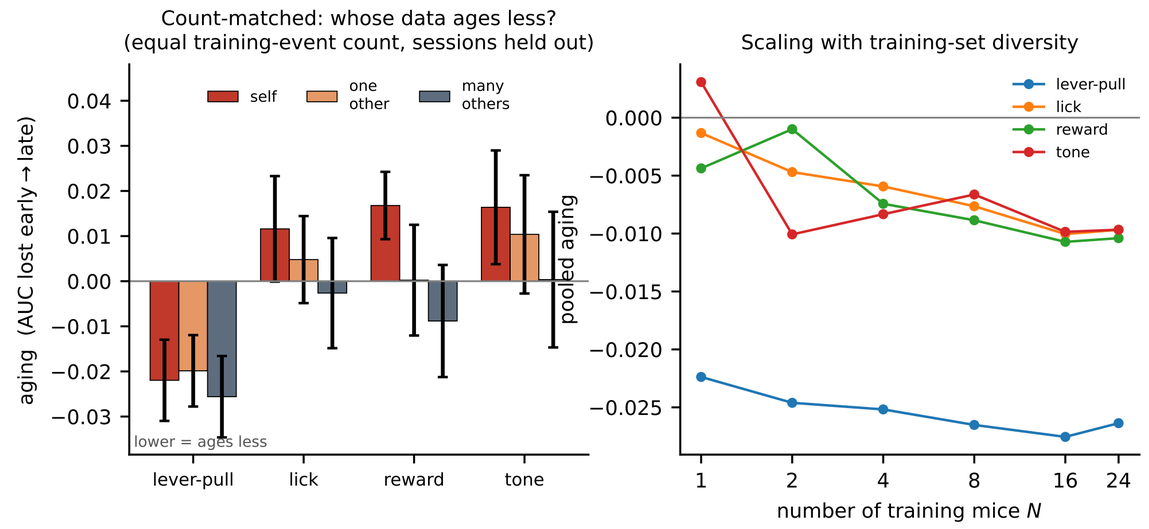
The pooled decoder *gains* over days as the animal becomes more practised (+0.0015/day) while the personal decoder stays flat — fit to idiosyncrasies, it cannot ride that wave, and the lift it fails to take is what opens the gap.

Takeaway

The early-week accuracy cost of decoding across individuals is **repaid as late-week stability**. Population priors are a credible alternative to per-subject recalibration — and they need no new labelled data from the subject at deployment time, which is the expensive part.

Not a data-volume artifact

A pooled decoder sees ∼24× more training events. So we fix the training-set size to n and compare three decoders scored on identical held-out data.



At **equal data**, the subject's own decoder still ages most (+0.010 AUC vs. one other mouse). Training on *many* others beats *one* other on all four events — diversity adds a further, smaller gain.

It replicates on an independent cohort

	Asymmetry (95% CI)	p	own ages more
BraiDyn-BC, widefield, 25 mice	$+0.017$ $[0.010, 0.024]$	5×10^{-5}	73%
Allen VB, two-photon, 23 mice	$+0.023$ $[0.010, 0.037]$	0.003	74%

Different laboratory, cellular two-photon instead of mesoscale widefield, a visual change-detection task instead of a cued lever pull, and a permutation-invariant population summary instead of an atlas feature. **Little is shared but the question and the protocol.**

The *mechanism* differs: on BraiDyn-BC the gap narrows because the pooled decoder gains; here because the personal decoder loses. The shared finding is that the personal decoder ages faster.

Measuring it correctly

Repeating the analysis with a **trial-level split** inflates the personal decoder's within-block accuracy by up to +0.013 AUC and leaves the pooled decoder untouched.

On lever pull that inflation is the **entire effect**: $+0.014$ ($p = 0.001$) → $+0.003$ ($p = 0.37$).

Any within- vs. cross-subject comparison is exposed to this. **Hold out sessions.**

Code & streamed-feature pipeline: `github.com/v1hns/braidyn-decoding`
Both cohorts are public: DANDI:001425 and the Allen Brain Observatory.

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